

ap 静态负载均衡

一、实验目的

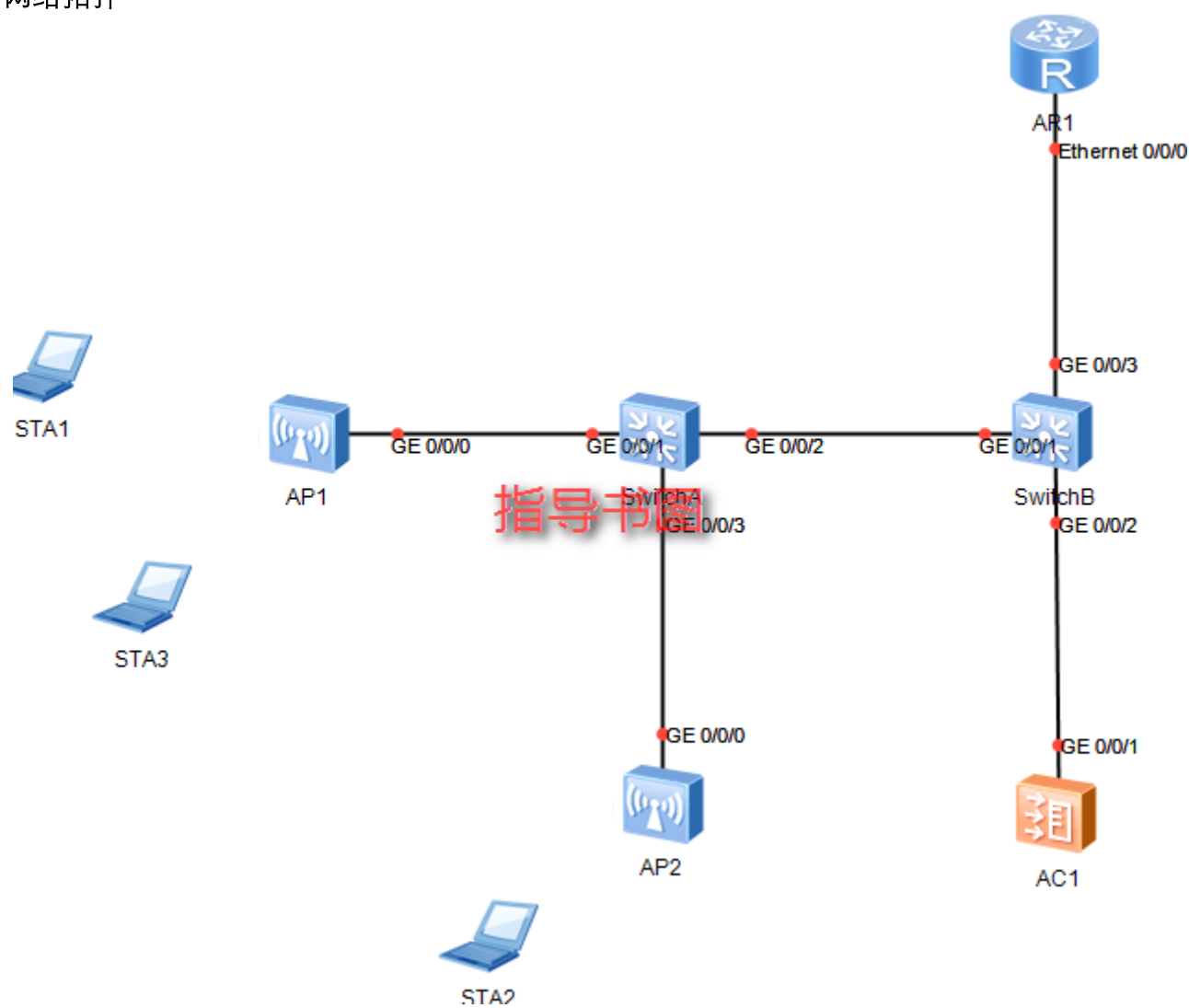
了解 ap 静态负载均衡相关知识、目的及配置方法。

二、模拟器版本

130

三、实验拓扑

3.1 网络拓扑



3.2 参数规划表

配置项	规划数据
管理 VLAN	VLAN10、VLAN100
STA 业务 VLAN	VLAN pool
	名称: sta-pool
	VLAN pool 中加入的 VLAN: 101、102
DHCP 服务器	AC 作为 AP 的 DHCP 服务器
	汇聚交换机作为 STA 的 DHCP 服务器, STA 的默认网关为 10.23.101.2 和 10.23.102.2

AP 的 IP 地址池	10.23.100.2 ~ 10.23.100.254/24
STA 的 IP 地址池	10.23.101.3 ~ 10.23.101.254/24
	10.23.102.3 ~ 10.23.102.254/24
AC 的源接口	IP 地址 VLANIF100: 10.23.100.1/24
AP 组	名称: ap1
	引用模板: VAP 模板 wlan-net、域管理模板 default
域管理模板	名称: default 国家码: 中国
SSID 模板	名称: wlan-net
	SSID 名称: wlan-net
安全模板	名称: wlan-net
	安全策略: WPA-WPA2+PSK+AES
	密码: a1234567
VAP 模板	名称: wlan-net
	转发模式: 直接转发
	业务 VLAN: VLAN pool
	引用模板: SSID 模板 wlan-net、安全模板 wlan-net
静态负载均衡组	名称: wlan-static
	负载均衡起始阈值: 10 个 负载均衡差值阈值: 5%

四、实验步骤

4.1 根据网络拓扑在模拟器中搭建网络;

4.2 步骤 1 配置周边设备

配置接入交换机SwitchA 的GE0/0/1、GE0/0/2 和GE0/0/3 接口加入VLAN10、VLAN101 和VLAN102, GE0/0/1 和GE0/0/3 的缺省VLAN 为VLAN10。

```
<HUAWEI> system-view
```

```
[HUAWEI] sysname SwitchA
```

```
[SwitchA] vlan batch 10 101 102
```

```
[SwitchA] interface gigabitethernet 0/0/1
```

```
[SwitchA-GigabitEthernet0/0/1] port link-type trunk
```

```
[SwitchA-GigabitEthernet0/0/1] port trunk pvid vlan 10
```

```
[SwitchA-GigabitEthernet0/0/1] port trunk allow-pass vlan 10 101 102
```

```
[SwitchA-GigabitEthernet0/0/1] port-isolate enable
```

```
[SwitchA-GigabitEthernet0/0/1] quit
```

```
[SwitchA] interface gigabitethernet 0/0/3
```

```
[SwitchA-GigabitEthernet0/0/3] port link-type trunk
```

```
[SwitchA-GigabitEthernet0/0/3] port trunk pvid vlan 10
```

```
[SwitchA-GigabitEthernet0/0/3] port trunk allow-pass vlan 10 101 102
```

```
[SwitchA-GigabitEthernet0/0/3] port-isolate enable
```

```
[SwitchA-GigabitEthernet0/0/3] quit
```

```
[SwitchA] interface gigabitethernet 0/0/2
```

```
[SwitchA-GigabitEthernet0/0/2] port link-type trunk
```

```
[SwitchA-GigabitEthernet0/0/2] port trunk allow-pass vlan 10 101 102
```

```
[SwitchA-GigabitEthernet0/0/2] quit
```

配置汇聚交换机SwitchB 的接口GE0/0/1 加入VLAN10、VLAN101 和VLAN102, 接口GE0/0/2 加入VLAN100, 接口GE0/0/3 加入VLAN101 和VLAN102, 并创建接口VLANIF100, 地址为

10.23.100.2/24。

```
<HUAWEI> system-view
[HUAWEI] sysname SwitchB
[SwitchB] vlan batch 10 100 101 102
[SwitchB] interface gigabitethernet 0/0/1
[SwitchB-GigabitEthernet0/0/1] port link-type trunk
[SwitchB-GigabitEthernet0/0/1] port trunk allow-pass vlan 10 101 102
[SwitchB-GigabitEthernet0/0/1] quit
[SwitchB] interface gigabitethernet 0/0/2
[SwitchB-GigabitEthernet0/0/2] port link-type trunk
[SwitchB-GigabitEthernet0/0/2] port trunk allow-pass vlan 100
[SwitchB-GigabitEthernet0/0/2] quit
[SwitchB] interface gigabitethernet 0/0/3
[SwitchB-GigabitEthernet0/0/3] port link-type trunk
[SwitchB-GigabitEthernet0/0/3] port trunk allow-pass vlan 101 102
[SwitchB-GigabitEthernet0/0/3] quit
[SwitchB] interface vlanif 100
[SwitchB-Vlanif100] ip address 10.23.100.2 24
[SwitchB-Vlanif100] quit
# 配置Router 的接口GE1/0/0 加入VLAN101 和VLAN102，创建接口VLANIF101 并配置IP 地址为10.23.101.2/24，创建接口VLANIF102 并配置IP 地址为10.23.102.2/24。
```

```
<Huawei> system-view
[Huawei] sysname Router
[Router] vlan batch 101 102
[Router] interface gigabitethernet 1/0/0
[Router-GigabitEthernet1/0/0] port link-type trunk
[Router-GigabitEthernet1/0/0] port trunk allow-pass vlan 101 102
[Router-GigabitEthernet1/0/0] quit
[Router] interface vlanif 101
[Router-Vlanif101] ip address 10.23.101.2 24
[Router-Vlanif101] quit
[Router] interface vlanif 102
[Router-Vlanif102] ip address 10.23.102.2 24
[Router-Vlanif102] quit
```

4.3 步骤 2 配置 AC 与其它网络设备互通

配置AC 的接口GE0/0/1 加入VLAN100，并创建接口VLANIF100。

```
<AC6605> system-view
[AC6605] sysname AC
[AC] vlan batch 100
[AC] interface vlanif 100
[AC-Vlanif100] ip address 10.23.100.1 24
[AC-Vlanif100] quit
[AC] interface gigabitethernet 0/0/1
[AC-GigabitEthernet0/0/1] port link-type trunk
[AC-GigabitEthernet0/0/1] port trunk allow-pass vlan 100
[AC-GigabitEthernet0/0/1] quit
```

配置AC 到AP 的路由，下一跳为SwitchB 的VLANIF100。

```
[AC] ip route-static 10.23.10.0 24 10.23.100.2
```

4.4 步骤 3 配置 DHCP 服务器为 STA 和 AP 分配 IP 地址

在SwitchB 上配置DHCP 中继，代理AC 分配IP 地址。

```
[SwitchB] dhcp enable
```

```
[SwitchB] ip pool vlan101
```

```
[SwitchB-ip-pool-vlan101] gateway-list 10.23.101.2
```

```
[SwitchB-ip-pool-vlan101] network 10.23.101.0 mask 255.255.255.0
```

```
[SwitchB-ip-pool-vlan101] quit
```

```
[SwitchB] ip pool vlan102
```

```
[SwitchB-ip-pool-vlan102] gateway-list 10.23.102.2
```

```
[SwitchB-ip-pool-vlan102] network 10.23.102.0 mask 255.255.255.0
```

```
[SwitchB-ip-pool-vlan102] quit
```

```
[SwitchB] interface vlanif 10
```

```
[SwitchB-Vlanif10] ip address 10.23.10.1 24
```

```
[SwitchB-Vlanif10] dhcp select relay
```

```
[SwitchB-Vlanif10] dhcp relay server-ip 10.23.100.1
```

```
[SwitchB-Vlanif10] quit
```

在SwitchB 上创建VLANIF101 和VLANIF102 接口为STA 提供地址，并指定默认网关。

```
[SwitchB] interface vlanif 101
```

```
[SwitchB-Vlanif101] ip address 10.23.101.1 24
```

```
[SwitchB-Vlanif101] dhcp select global
```

```
[SwitchB-Vlanif101] quit
```

```
[SwitchB] interface vlanif 102
```

```
[SwitchB-Vlanif102] ip address 10.23.102.1 24
```

```
[SwitchB-Vlanif102] dhcp select global
```

```
[SwitchB-Vlanif102] quit
```

在AC 上创建全局地址池为AP 提供地址。

```
[AC] dhcp enable
```

```
[AC] ip pool huawei
```

```
[AC-ip-pool-huawei] network 10.23.10.0 mask 24
```

```
[AC-ip-pool-huawei] gateway-list 10.23.10.1
```

```
[AC-ip-pool-huawei] option 43 sub-option 3 ascii 10.23.100.1
```

```
[AC-ip-pool-huawei] quit
```

```
[AC] interface vlanif 100
```

```
[AC-Vlanif100] dhcp select global
```

```
[AC-Vlanif100] quit
```

4.5 步骤 4 配置 VLAN pool，用于作为业务 VLAN

在AC 上新建VLAN pool，并将VLAN101 和VLAN102 加入其中，配置VLAN pool 中的VLAN 分配算法为“hash”。

```
[AC] vlan pool sta-pool
```

```
[AC-vlan-pool-sta-pool] vlan 101 102
```

```
[AC-vlan-pool-sta-pool] assignment hash
```

```
[AC-vlan-pool-sta-pool] quit
```

4.6 步骤 5 配置 AP 上线

创建AP 组，用于将相同配置的AP 都加入同一AP 组中。

```
[AC] wlan
```

```
[AC-wlan-view] ap-group name ap1
```

```
[AC-wlan-ap-group-ap1] quit
# 创建域管理模板，在域管理模板下配置AC 的国家码并在AP 组下引用域管理模板。
[AC-wlan-view] regulatory-domain-profile name default
[AC-wlan-regulate-domain-default] country-code cn
[AC-wlan-regulate-domain-default] quit
[AC-wlan-view] ap-group name ap1
[AC-wlan-ap-group-ap1] regulatory-domain-profile default
Warning: Modifying the country code will clear channel, power and antenna gain configurations
of the radio and reset the AP. Continue?[Y/N]:y
[AC-wlan-ap-group-ap1] quit
[AC-wlan-view] quit
# 配置AC 的源接口。
[AC] capwap source interface vlanif 100
# 在AC 上离线导入AP，并将AP 加入AP 组“ap1”中
[AC] wlan
[AC-wlan-view] ap auth-mode mac-auth
[AC-wlan-view] ap-id 0 ap-mac 00e0-fc59-72d0
[AC-wlan-ap-0] ap-name area_1
[AC-wlan-ap-0] ap-group ap1
Warning: This operation may cause AP reset. If the country code changes, it will clear channel,
power and antenna gain configurations of the radio, Whether to continue? [Y/N]:y
[AC-wlan-ap-0] quit
[AC-wlan-view] ap-id 1 ap-mac 00e0-fc13-7ed0
[AC-wlan-ap-1] ap-name area_2
[AC-wlan-ap-1] ap-group ap1
Warning: This operation may cause AP reset. If the country code changes, it will clear channel,
power and antenna gain configurations of the radio, Whether to continue? [Y/N]:y
[AC-wlan-ap-1] quit
# 将AP 上电后，当执行命令display ap all 查看到AP 的“State”字段为“nor”时，表示AP
正常上线。
```

```
<AC>display ap all
Info: This operation may take a few seconds. Please wait for a moment.done.
Total AP information:
nor : normal          [2]
-----
ID   MAC                Name   Group IP      Type           State STA Uptime
-----
0    00e0-fc59-72d0 area_1 ap1    10.23.10.13 AP4050DN-E     nor    0    2S
1    00e0-fc13-7ed0 area_2 ap1    10.23.10.239 AP4050DN-E     nor    0    12S
-----
Total: 2
```

4.7 步骤6 配置 WLAN 业务参数

```
# 创建名为“wlan-net”的安全模板，并配置安全策略。
[AC-wlan-view] security-profile name wlan-net
[AC-wlan-sec-prof-wlan-net] security wpa-wpa2 psk pass-phrase a1234567 aes
[AC-wlan-sec-prof-wlan-net] quit
# 创建名为“wlan-net”的SSID 模板，并配置SSID 名称为“wlan-net”。
```

```

[AC-wlan-view] ssid-profile name wlan-net
[AC-wlan-ssid-prof-wlan-net] ssid wlan-net
[AC-wlan-ssid-prof-wlan-net] quit
# 创建名为“wlan-net”的VAP 模板，配置业务数据转发模式、业务VLAN，并且引用安全模板和SSID 模板。
[AC-wlan-view] vap-profile name wlan-net
[AC-wlan-vap-prof-wlan-net] forward-mode direct-forward
[AC-wlan-vap-prof-wlan-net] service-vlan vlan-pool sta-pool
[AC-wlan-vap-prof-wlan-net] security-profile wlan-net
[AC-wlan-vap-prof-wlan-net] ssid-profile wlan-net
[AC-wlan-vap-prof-wlan-net] quit
# 配置AP 组引用VAP 模板，AP 上射频0 和射频1 都使用VAP 模板“wlan-net”的配置。
[AC-wlan-view] ap-group name ap1
[AC-wlan-ap-group-ap1] vap-profile wlan-net wlan 1 radio 0
[AC-wlan-ap-group-ap1] vap-profile wlan-net wlan 1 radio 1
[AC-wlan-ap-group-ap1] quit

```

4.8 步骤 7 配置 AP 射频的信道和功率

关闭射频的信道和功率自动调优功能。射频的信道和功率自动调优功能默认开启，如果不关闭此功能则会导致手动配置不生效。

```

[AC-wlan-view] rrm-profile name default
[AC-wlan-rrm-prof-default] calibrate auto-channel-select disable
[AC-wlan-rrm-prof-default] calibrate auto-txpower-select disable
[AC-wlan-rrm-prof-default] quit
# 配置AP 射频0 的信道和功率。
[AC-wlan-view] ap-id 0
[AC-wlan-ap-0] radio 0
[AC-wlan-radio-0/0] channel 20mhz 6
Warning: This action may cause service interruption. Continue?[Y/N]y
[AC-wlan-radio-0/0] eirp 127
[AC-wlan-radio-0/0] quit
# 配置AP 射频1 的信道和功率。
[AC-wlan-ap-0] radio 1
[AC-wlan-radio-0/1] channel 20mhz 149
Warning: This action may cause service interruption. Continue?[Y/N]y
[AC-wlan-radio-0/1] eirp 127
[AC-wlan-radio-0/1] quit
[AC-wlan-ap-0] quit

```

4.9 步骤 8 配置静态负载均衡功能

#创建静态负载均衡组，将AP area_1 和area_2 加入静态负载均衡组。

```

[AC-wlan-view] sta-load-balance static-group name wlan-static
[AC-wlan-sta-lb-static-wlan-static] member ap-name area_1
[AC-wlan-sta-lb-static-wlan-static] member ap-name area_2
#指定静态负载均衡起始阈值为10 个，负载均衡差值阈值为5%。
[AC-wlan-sta-lb-static-wlan-static] start-threshold 10
[AC-wlan-sta-lb-static-wlan-static] gap-threshold 5
[AC-wlan-sta-lb-static-wlan-static] quit

```

4.10 步骤 9 验证配置结果

在AC 上执行命令display sta-load-balance static-group name wlan-static，可以查看到静态负载均衡的相关信息。

```
<AC>display sta-load-balance static-group name wlan-static
-----
Group name           : wlan-static
Load-balance status  : balance
Start threshold      : 10
Gap threshold(%)     : 5
Deny threshold      : 3
-----
RfID: Radio ID
CurEIRP: Current EIRP (dBm)
Act CH: Actual channel, Cfg CH: Config Channel
-----
AP ID  AP Name  RfID  Act CH/Cfg CH  CurEIRP/MaxEIRP  Client
-----
0      area_1  0      6/6             -/-              0
0      area_1  1     149/149         -/-              0
1      area_2  0      1/-             -/-              0
1      area_2  1     149/-           -/-              0
-----
Total: 4
<AC>
```